carrying. During which phase is this most likely to occur? **e. Prophase I**

22 Following is a summary of what should appear in your drawings and descriptions of the stages of meiosis. For further reference, check out Figure 16-3.

In the drawing for late prophase I, at least two pairs of homologous chromosomes should be grouped into tetrads. (In truth, there are 23 pairs, but simplified illustrations tend to show just two.) The description for prophase I should include reference to the tetrad formation. The drawing for metaphase I should show the equatorial plane (a center horizontal line) with the tetrads aligned along it; an “x” on each side. The illustration also should show spindles radiating from each pole, with the tetrads attached to them by their centromeres. The description should include reference to the equatorial plane, the poles, and the spindles.

The drawing for anaphase I should show the tetrads moving to the top and bottom of the cell along the spindles and the cytoplasm slowly beginning to divide. In telophase I, the division becomes more pronounced and two new nuclei form. The chromosomes remain in the “x” shape. As the process enters interkinesis, the cytoplasm pinches off into two cells.

During prophase II, which also is the start of the second meiotic division, the chromosomes migrate toward a new equatorial plane. The drawing of metaphase II should show all chromosomes aligned on the equatorial plane with the centromere along the (invisible) center line. For anaphase II, you should show the chromosomes pulling apart into chromatids and moving toward the poles. In the final stage, telophase II, you should draw new nuclei forming around the chromatin as it uncoils at this point.

Chapter 17

Carrying Life: The Female Reproductive System

IN THIS CHAPTER

- **Mapping out the female reproductive parts and what they do**
- **Understanding meiosis as the process that makes eggs**
- **Explaining embryology**
- **Nursing a fetus into a baby**
- **Following the process of growth and aging in women**

Men may have quite a few hard-working parts in their reproductive systems (see Chapter 16), but women are the ones truly responsible for survival of the species (biologically speaking, anyway). The female body prepares for reproduction every month for most of a woman's adult life, producing a secondary oocyte and then measuring out delicate levels of hormones to prepare for nurturing a developing embryo. When a fertilized ovum fails to show up, the body hits the biological reset button and sloughs off the uterine lining before building it up all over again for next month's reproductive roulette. But that's nothing compared to what the female body does when a fertilized egg actually settles in for a nine-month stay. Strap yourselves in for a tour of the incredible female baby-making machinery — practice questions included.

Identifying the Female Reproductive Parts and Their Functions

Before we dive in to major reproductive function, let's take a brief tour of the female reproductive system's parts.

External genitalia

Unlike in the male, most of a female's reproductive organs are internal. However, there are external ones that play a role. Together, the external structures are referred to as the *vulva*. These include:

- **Mons pubis:** Fatty tissue that lies atop the pubic symphysis, secretes pheromones, and grows hair to trap particles, preventing them from entering the reproductive tract
- **Labia majora:** The outer folds of tissue that protect the area, contain sweat and sebaceous (oil) glands to aid in lubrication, homologous to the scrotum
- **Labia minora:** Surround the openings to the urethra and vagina, are highly vascularized and become engorged during arousal to increase sensitivity (to pleasurable sensation)
- **Clitoris:** A small mound of erectile tissue on the anterior end of where there the labia minora meet, it is the most sensitive area of the female genitalia — repeated stimulation can lead to orgasm; homologous to the penis
- **Bartholin glands:** Located at the opening of the vagina, secretes a thick fluid for lubrication during intercourse

Internal genitalia

The remaining organs are all found internally in the reproductive tract. These include:

- **Vagina:** Opens to the external genitalia, receives penis for intercourse; serves as *birth canal* for childbirth
- **Cervix:** A ring of muscular and connective tissue at the end of the vagina, guarding the uterus, produces a thick mucus to trap particles and bacteria except before ovulation where it thins to allow passage for sperm
- **Uterus:** Pear-shaped organ for housing a developing fetus, the lining, or *endometrium*, is triggered by estrogen to add bulk early in a woman's cycle to prepare it for the embryo; if fertilization does not occur, the lining is shed during *menstruation* and the process starts over

- **Uterine tubes (also known as *Fallopian tubes* or *oviducts*):** Attached at the top right and left sides of the uterus, widen at the ovary ends into the fan-shaped *infundibulum*, site of fertilization
- **Ovaries:** The almond-shaped gonads, produce estrogen and progesterone, develop eggs

TIP The term "estrogen" actually refers to a group of hormones primarily produced in the ovaries. These are responsible for developing the reproductive organs as well as providing the female secondary sex characteristics (as testosterone does in males). The primary estrogen hormone is *estradiol*, so when estrogen is used generically, this is the hormone being referred to.

Gamete production

First and foremost, in the female reproductive repertoire are the two *ovaries*, which usually take a turn every other month to produce a single *ovum* or *secondary oocyte*. A sheet of connective tissue called the broad ligament holds the ovaries in place. Each ovary is surrounded by a dense fibrous connective tissue called the *tunica albuginea* (literally "white covering"); yes, that's the same name as the tissue surrounding the testes. In fact, the ovaries in a female and the testes in a male are *homologous*, meaning that they share similar origins. The ovaries themselves are arranged into an outer layer (the cortex) and an inner layer (the medulla). The cortex houses the *primordial follicles*, which will eventually mature into ova. The medulla is loosely packed with blood and lymphatic vessels and nerves.

Ovarian cycle

During fetal development, cells in the ovaries called *oogonia* begin meiosis, creating about two million *primary oocytes* suspended in prophase I. By puberty, though, about three-quarters of them have died. At the onset of puberty, the primary oocytes resume meiosis, splitting unevenly into a *polar body* and the *secondary oocyte*. These are wrapped in a single layer of epithelial tissue creating the *primary follicle*. They continue meiosis and the polar body (which is just a nucleus in a cell membrane) completes the process creating two polar bodies which will just degenerate. The secondary oocyte pauses meiosis again, this time at metaphase II. It will not complete meiosis until fertilization has been initiated — finally creating the actual ovum and a polar body that will degenerate.

REMEMBER At puberty and approximately once each month until menopause (which we cover in the later section “Growing, Changing, and Aging”), the *ovarian cycle* occurs as follows:

1. The anterior pituitary gland secretes *follicle-stimulating hormone,* or *FSH,* which prompt about 1,000 of the primordial follicles to resume meiosis.

   Usually, one follicle outgrows the rest, maturing into a *Graafian follicle.* Non-identical, or *fraternal,* twins, triplets, or even more embryos result if more than one follicle matures and more than one secondary oocyte is released (and fertilized).

2. Follicular cells divide and surround the oocyte with several layers of cells called the *cumulus oophorus*, creating the *secondary follicle*.

3. As maturation continues, the cell layers separate leaving a fluid-filled space between them. The cells in contact with the oocyte form the *zona pelludica*. The others secrete estrogen to signal the uterus to prepare by thickening the endometrium.

4. As blood levels of estrogen begin to rise, the pituitary stops releasing FSH and begins releasing *luteinizing hormone,* or *LH,* which prompts the Graafian follicle now at the surface of the ovary to rupture, triggering *ovulation* — the release of the secondary oocyte, more commonly referred to as an egg cell.

5. The oocyte, wrapped in the zona pelludica, takes some follicle cells with it when released, which form the *corona radiata*. They serve to provide nourishment for the journey. Since the uterine tube is not directly connected to the ovary, the egg is released into the pelvic cavity. However, the end of the uterine tube, the infundibulum, is crowned in finger-like projections called *fimbriae*. Cells of this tissue are ciliated, which wave at ovulation to usher the egg into the tube.

6. It takes approximately three days for the egg to travel through the uterine tube to the uterus, regardless of whether or not it is fertilized. The egg is only viable, however, for about 24 hours.

Meanwhile, back at the ovary, a clot has formed inside the ruptured follicle, and the remaining follicle cells form the *corpus luteum* (literally “yellow body”). This new but temporary endocrine gland secretes *progesterone*, a hormone that signals the uterine lining to prepare for possible implantation of a fertilized egg; inhibits the maturing of follicles, ovulation, and the production of estrogen to prevent menstruation; and stimulates further growth in the mammary glands (which is why some women get sore breasts a few days before their periods begin). If pregnancy occurs, the placenta also will release progesterone to prevent menstruation throughout the pregnancy.

If the egg is not fertilized, the corpus luteum dissolves after 10 to 14 days to be replaced by scar tissue called the *corpus albicans*. If pregnancy does occur, the corpus luteum remains and grows for about six months before disintegrating. Only about 400 of a woman’s primordial follicles ever develop into secondary oocytes for the trip to the uterus. Figure 17-1 shows what happens inside an ovary.

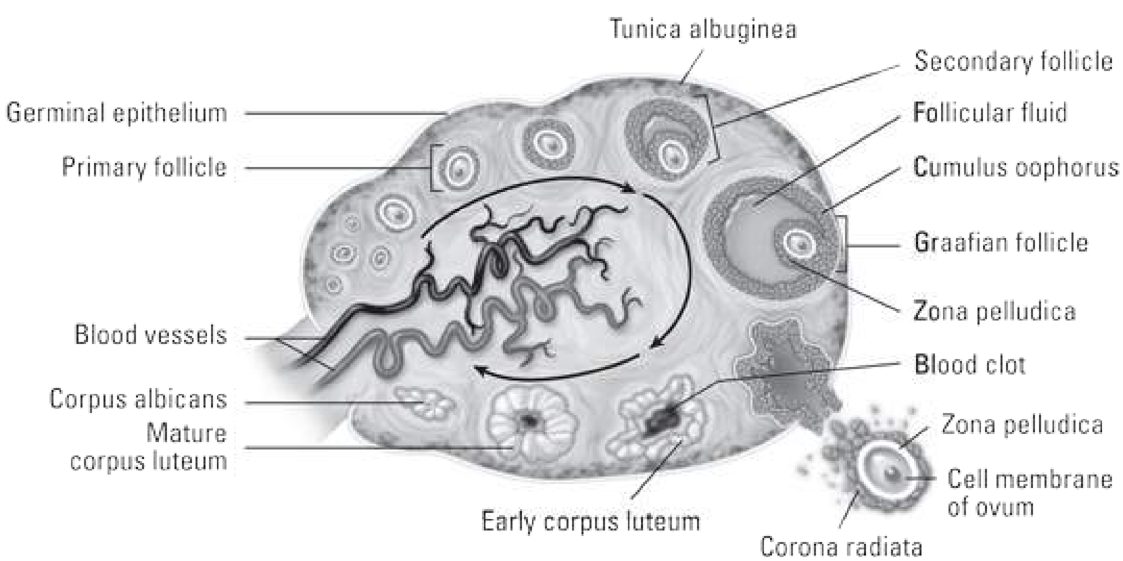

Illustration by Imagineering Media Services Inc.

FIGURE 17-1: The ovary.

Uterine cycle

Occurring alongside the ovarian cycle is the *uterine cycle*, which is directly influenced by the hormones made in the ovaries. Day one of the cycle occurs at the onset of menstruation, when a woman “starts her period” so to speak. Because the egg was not fertilized, the uterus must hit reset (the *menstrual phase*) and shed the endometrium; otherwise it would not be hospitable for the new embryo when it arrives following the next ovarian cycle. After the ovaries start releasing estrogen, the uterus begins its *proliferation phase* by creating more connective tissue, thickening the endometrium. Following ovulation and the release of progesterone, the uterus enters the *secretory phase* and increases blood flow to the endometrium in order to provide nutrients for the embryo. (For more on this process, flip ahead to the “Making Babies: An Introduction to Embryology” section.)

The middle layer of the uterine wall, the *myometrium*, is comprised of smooth muscle. It contracts during menstruation to slough off the inner lining. If pregnancy is achieved, progesterone blocks any uterine contraction until the onset of labor. At that point, the uterus begins contracting with increasing frequency and force to push the fetus through the birth canal.

Find out how familiar you are with the female reproductive structures and their functions:

1 True or False: The medulla of the ovary is the site of follicular development.

2-11 Fill in the blanks to complete the paragraph.

The **2.** _________ in a developing female fetus begin meiosis. Upwards of 2 million **3.**_________________ are produced but they are suspended in **4.**___________. By the onset of **5.**_________ only about 300,000–400,000 remain. Each month, about 1,000 primary follicles are triggered to resume meiosis by **6.**_______. One usually outgrows the rest, forming the **7.**___________________. Meiosis is again suspended, this time at **8.**_________________. Release of the secondary oocyte, or **9.**_____________, is triggered by the release of **10.**________ from the anterior pituitary. Meiosis II is not completed until **11.**_____________ has begun.

12-16 Match the term to its description.

a. Fingerlike projections at the end of a uterine tube

b. Protective layer directly surrounding the secondary oocyte

c. Granulosa cells forming the outer layer of the Graafian follicle

d. Layer of smooth muscle in uterine wall

e. Inner lining of the uterus

12. ______ Corona radiata
13. ______ Endometrium
14. ______ Fimbriae
15. ______ Zona pelludica
16. ______ Myometrium

17 Which one of the following is NOT a function of estradiol?

a. Preparing the endometrium

b. Supporting development of the secondary oocyte

c. Supporting development of the corpus luteum

d. Triggering development of the reproductive organs

e. Preventing secretion of FSH from the pituitary

18-33 Use the terms that follow to identify the anatomy of the female reproductive system shown in Figure 17-2.

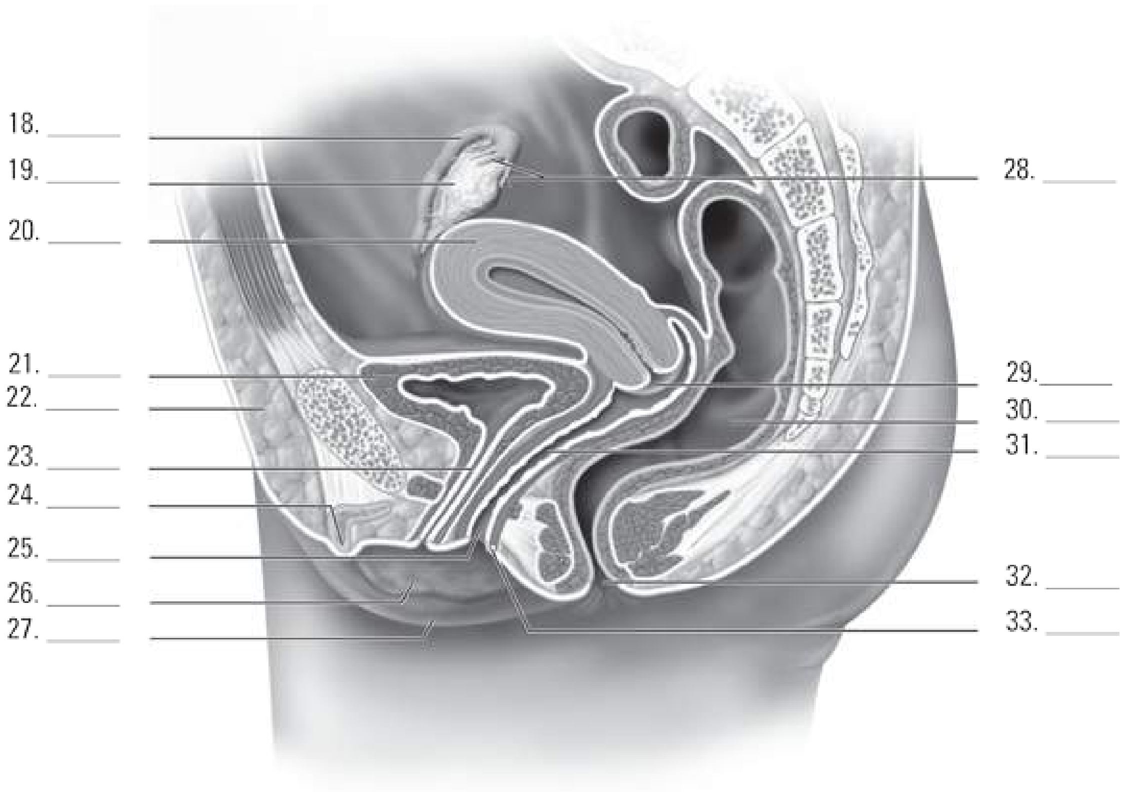

Illustration by Imagineering Media Services Inc.

FIGURE 17-2: The female reproductive system.

a. Cervix

b. Fimbriae

c. Urinary bladder

d. Clitoris

e. Labium major

f. Ovary

g. Rectum

h. Bartholin’s glands

i. Vaginal orifice

j. Anus

k. Mons pubis

l. Uterus

m. Labium minor

n. Uterine (Fallopian) tube

o. Vagina

p. Urethra

34 Why does the corpus luteum produce progesterone?

a. To stimulate development of the cumulus oophorus

b. To trigger ovulation

c. To prepare a woman's uterus for pregnancy and prevent menstruation

d. To trigger the resumption of meiosis

e. To prepare the infundibulum to fragment into fimbriae

35 True or False: Once a sperm penetrates the cell membrane of the ovum, the zona pelludica immediately hardens.

Making Eggs: A Mite More Meiosis

We cover the details of meiosis in Chapter 16, but in this section we explore how meiosis contributes to creating a *zygote* (fertilized egg) and its chromosomal makeup. Normal human diploid, or 2N, cells contain 23 pairs of homologous chromosomes for a total of 46 chromosomes each. One chromosome from each pair comes from the individual's mother, and the other comes from the father. Each homologous pair contains the same type of genetic information, but the *allele* (variety) of this genetic information may differ from one chromosome of the pair to the homologous one. For example, one chromosome from the mother may carry the genetic coding for blue eyes, whereas the homologous chromosome from the father may code for brown eyes.

REMEMBER Although all the body's cells have 46 chromosomes — including the cells that eventually mature into the ovum and the sperm — each gamete (either ovum or sperm) must offer only half that number if fertilization is to succeed. That's where meiosis steps in. As you see in Figure 17-3, the number of chromosomes is cut in half during the first meiotic division, producing gametes that are haploid, or 1N. They, however, still contain two alleles for each gene so another division must occur. The second meiotic division produces four haploid sperms in the male but only one functional haploid secondary oocyte in the female. When fertilization occurs, the new zygote will contain 23 pairs of homologous chromosomes for a total of 46 chromosomes. The zygote then proceeds through mitosis to produce the body’s cells, distributing copies of all 46 chromosomes to each new cell.

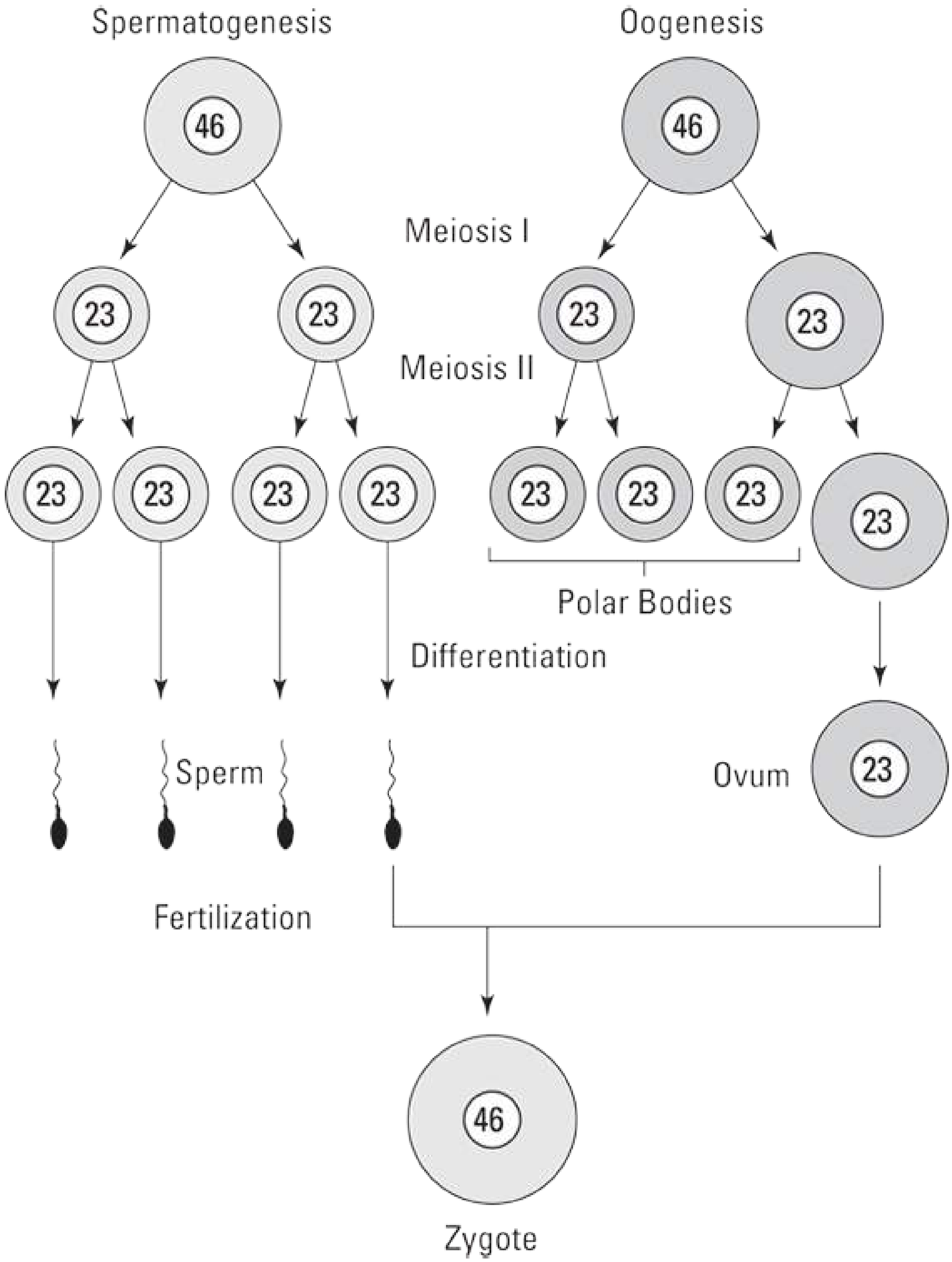

FIGURE 17-3: How chromosomes divide up in meiosis.

Think you've conquered this process? Find out by answering the following practice questions:

EXAMPLE **Q.** How many chromosomes are in most of the cells in your body?

a. 46

b. 23

c. 92

A. The correct answer is 46. That's the usual human complement.

36-44 Fill in the blanks to complete the following sentences:

Meiosis produces sperm and ova, which when combined make a **36.**__________ (fertilized egg) with its full complement of chromosomes. Normally, humans have **37.**_________________, or 2N, cells containing 23 pairs of homologous chromosomes for a total of 46 chromosomes each. A pair of chromosomes containing the same type of genetic information are **38.**_________________ chromosomes though they may have different **39.**_________________ or varieties of a gene. Ova and sperm are called sex cells or **40.**_________________. The number of chromosomes is cut in half during the first meiotic division, producing gametes that are **41.**_________________, or 1N. The second meiotic division produces **42.**_________________ sperm in the male but only **43.**_________________ secondary oocyte in the female. The zygote then proceeds through **44.**_________________ to produce the body's cells.

Making Babies: An Introduction to Embryology

Fertilization (the penetration of a secondary oocyte by a sperm) must occur in the uterine tube within 24 hours of ovulation or the secondary oocyte degenerates. But that doesn't mean that sexual intercourse outside that time frame can't lead to pregnancy. Research indicates that spermatozoa can survive up to seven days inside the female reproductive tract. If a sperm is still motile — that is, if it's still whipping its flagellum tail — when an ovum comes down the tube, it will do what it was made to do and penetrate the ovum’s membrane.

The sperm releases enzymes that allow it to digest its way into the secondary oocyte, leaving its flagellum behind. With the penetration of the sperm, meiosis II is completed, resulting in an ovum and a second polar body. When the *pronuclei* of the ovum and the sperm unite, presto! You’ve got a zygote (fertilized egg). Over the next three to five days, the zygote moves through the uterine tube to the uterus, undergoing *cleavage* (mitotic cell division) along the way: Two cells become four smaller cells, four cells become eight smaller cells, and then those eight cells become a solid 16-cell ball called a *morula.* After five days of cleavage, the cells form a hollow ball called a *blastula,* or *blastocyst.* The inner hollow region is called the *blastocoele;* the outer-layer cells are called the *trophoblast;* and the inner mass of cells is the *embryoblast.* Figure 17-4 illustrates this process of development.

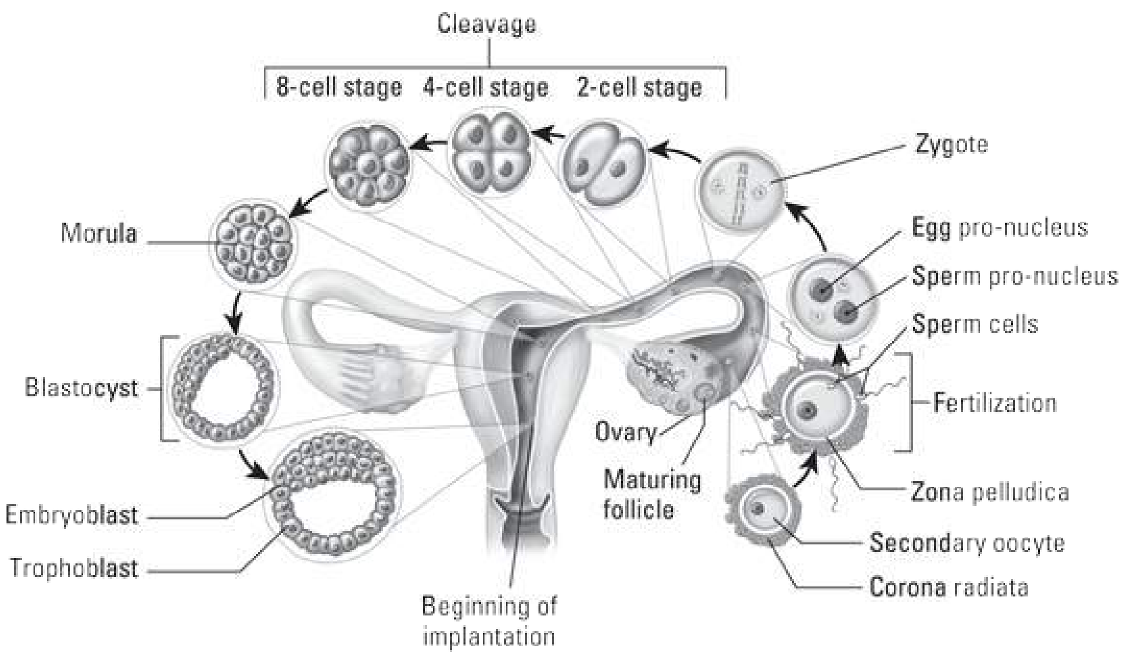

Illustration by Imagineering Media Services Inc.

FIGURE 17-4: Embryonic development.

Within three days of its arrival in the uterus (generally within a week of fertilization), the blastocyst implants in the endometrium, and some of the blastocyst’s cells — called *totipotent embryonic stem cells* — organize into an inner cell mass also known as the *embryonic disk.* (Refer to Chapter 3 for more on stem cells.) Over time, the embryonic disk *differentiates* into the tissues of the developing embryo. The outer cells of the disk form the *amnion* and its resulting *amniotic cavity*, and the inner ones form two primitive *germ layers*. The layer nearest the amniotic cavity forms the *ectoderm*, while the others form the *endoderm*. Between the two layers, cells differentiate to form a third layer, the *mesoderm*. Cells of the three germ layers develop into the body parts:

- **Ectoderm:** Skin, nervous tissue
- **Mesoderm:** Bones, cartilage, connective tissue, muscle
- **Endoderm:** Respiratory and digestive tract organs, glands

TIP To keep these terms straight, remember that *endo*– means "inside or within," *ecto*– means "outer or external," and *meso*– means "middle."

After three weeks of development, the heart begins beating. In the fourth week of development, the embryonic disk forms an elongated structure that attaches to the developing placenta by a connecting stalk. A head and jaws form while primitive buds sprout; the buds will develop into arms and legs. During the fifth through seventh weeks, the head grows rapidly and a face begins to form (eyes, nose, and a mouth). Fingers and toes grow at the ends of the elongating limb buds. All internal organs have started to form. After eight weeks of development, the embryo begins to have a more human appearance and is referred to as a *fetus*.

The outer cells of the blastocyst, together with the endometrium of the uterus, form the *placenta*, a new internal organ that exists only during pregnancy. The placenta attaches the fetus to the uterine wall, exchanges gases and waste between the maternal and fetal bloodstreams, and secretes hormones to sustain the pregnancy.

Now that you've refreshed your memory a bit about how babies are made (beyond the birds and bees talks), try the following practice questions:

45-53 Mark the statement with a T if it's true or an F if it's false:

45. _____ The mesoderm will form nervous tissue and skin.
46. _____ The embryonic stage is complete at the end of the eighth week.

47. _____ Outer cells of the embryonic disk form the germ layers.
48. _____ During the fifth through seventh weeks, the arm and leg buds elongate, and fingers and toes begin to form.
49. _____ Cleavage is successive mitotic divisions of the embryonic cells into smaller and smaller cells.
50. _____ As the zygote moves through the Fallopian tube, it undergoes meiosis.
51. _____ The placenta exchanges gases and waste between the maternal blood and the fetal blood.
52. _____ After five days of cleavage, the cells form into a hollow ball called the morula.
53. _____ Sexual intercourse five days before ovulation cannot lead to pregnancy.

Growing from Fetus to Baby

Pregnancy is divided into three periods called *trimesters* (although many new parents bemoan a postnatal fourth trimester until the baby sleeps through the night). The first 12 weeks of development mark the first trimester, during which *organogenesis* (organ formation) is established. During the second trimester (typically considered to be weeks 13 to 27), all fetal systems continue to develop and rapid growth triples the fetus's length. By the third trimester (typically considered to be weeks 28 to 40), all organ systems are functional and the fetus usually is considered *viable* (capable of surviving outside the womb) even if it's born prematurely. The overall growth rate slows in the third trimester, but the fetus gains weight rapidly.

REMEMBER Milestones in fetal development can be marked monthly.

- At the **end of the second month** (when the terminology changes from "embryo" to "fetus"), the head remains overly large compared to the developing body, and the limbs are still short. All major regions of the brain have formed.

- During the **third month,** body growth accelerates and head growth slows. The arms reach the length they will maintain during fetal development. The bones begin to ossify, and all body systems have begun to form. The circulatory (cardiovascular) system supplies blood to all the developing extremities, and even the lungs begin to practice "breathing" amniotic fluid. By the end of the third month, the external genitalia are visible in the male (ultrasound technicians call this a "turtle sign"). The fetus is a bit less than 4 inches long and weighs about 1 ounce.
- The body grows rapidly during the **fourth month** as legs lengthen, and the skeleton continues to ossify as joints begin to form. The face looks more human. The fetus is about 7 inches long and weighs 4 ounces.
- Growth slows during the **fifth month,** and the legs reach their final fetal proportions. Skeletal muscles become so active that the mother can feel fetal movement. Hair grows on the head, and *lanugo,* a profusion of fine, soft hair, covers the skin. The fetus is about 12 inches long and weighs ½ to 1 pound.
- The fetus gains weight during the **sixth month,** and eyebrows and eyelashes form. The skin is wrinkled, translucent, and reddish because of dermal blood vessels. The fetus is between 11 and 14 inches long and weighs a bit less than 1½ pounds.
- During the **seventh month,** skin becomes smoother as the fetus gains subcutaneous fat tissue. The eyelids, which are fused during the sixth month, open. Usually, the fetus turns to an upside-down position. It's between 13 and 17 inches long and weighs from 2½ to 3 pounds.
- During the **eighth month,** subcutaneous fat increases, and the fetus shows more baby-like proportions. The testes of a male fetus descend into the scrotum. The fetus is now 16 to 18 inches long and has grown to just under 5 pounds.
- During the **ninth month,** the fetus plumps up considerably with additional subcutaneous fat. Much of the lanugo is shed, and fingernails extend all the way to the tips of the fingers. The average newborn at the end of the ninth month is 20 inches long and weighs about 7½ pounds.

The following practice questions deal with the development of the fetus during its 40 weeks in the womb:

54 What needs to happen before a fetus can be considered viable?

a. The heart rate must be easily detected.

b. All organ systems must be functional.

c. Subcutaneous fat must be built up to at least an inch.

d. The brain is fully formed.

e. The fetus must have turned into an upside-down position.

55 Describe one new fetal development for each month:

3rd month: ______________________

4th month: ______________________

5th month: ______________________

6th month: ______________________

7th month: ______________________

8th month: ______________________

9th month: ______________________

Parturition

Near the end of the fetus' stay in the womb, the woman's body begins preparations for *parturition*, or childbirth. Progesterone steadily drops as the big day nears leading to mild contractions that most women don't even feel. The cervix begins to thin (called *effacement*) and dilate to allow the fetus into the birth canal (vagina). But the trigger to set off the cascade of hormones that maintain the process of *labor* still remains a mystery. It is widely believed that it is initiated by the fetus. Regardless of the trigger, parturition is a classic example of a *positive feedback loop*.

Prostaglandins cause early uterine contractions, which leads to stretch (in the non-contracting parts) of the uterus. This feeds back to the hypothalamus, causing it to release *oxytocin* from the posterior pituitary. This triggers uterine contractions, which causes more stretch, which causes more oxytocin and more contractions, and so on, until the baby is delivered.

Labor progresses through three stages, the first of which is the longest. Stage one is split into three phases (that any woman who has experienced them would prefer to forget):

» **Phase 1 — latent labor:** Contractions every 5–20 minutes, cervix dilates up to 5–6 cm, typically lasts from 6–12 hours

» **Phase 2 — active labor:** Increasingly stronger contractions every 3–5 minutes that last about 60 seconds, cervix dilates up to 8 cm, typically lasts from 3–5 hours

» **Phase 3 — transition:** Intense contractions every 2–3 minutes for 60–90 seconds, cervix fully dilates, typically lasts from 30–60 minutes — most women say this is the most painful part of the entire childbirth process

Transition prepares the body for the second stage of labor as contractions shift from an all-around squeeze to a downward push. A woman feels the urge to push with each contraction, which slow down to every 3–5 minutes and last for 60–75 seconds. It takes, on average, 20 minutes to 2 hours to complete this stage by delivering the baby. After a brief reprieve, weaker contractions resume to deliver the placenta. This final stage of labor generally take less than 30 minutes.

The hours following parturition are critical for the bond between mother and child. *Prolactin* begins triggering milk production in the mammary glands, and the nursing infant stimulates the release of oxytocin to release the milk, called *let down*. Oxytocin is also known as the "love hormone"; it's the neurotransmitter that provides the love sensation we feel for family.

Are you feeling the love? Let's find out by answering some questions:

56 Which hormones play an active role in the progression of labor?

I. Progesterone

II. Prostaglandins

III. Oxytocin

a. I only

b. III only

c. I and II

d. II and III

e. I, II, and III

57 Which stage of labor usually lasts the longest?

a. stage 1, phase 1

b. stage 1, phase 2

c. stage 1, phase 3

d. stage 2

e. stage 3

58 Explain how childbirth is an example of a positive feedback loop.

____

____

____

Growing, Changing, and Aging

From birth to 4 weeks of age, the newborn is called a *neonate*. Faced with survival after its physical separation from the mother, a neonate must abruptly begin to process food, excrete waste, obtain oxygen, and make circulatory adjustments.

From 4 weeks to 2 years of age, the baby is called an *infant*. Growth during this period is explosive under the stimulation of *growth hormone* from the pituitary gland, adrenal steroids, and thyroid hormones. The infant's *deciduous* teeth, also called *baby* or *milk teeth*, begin to form and erupt through the gums. The nervous system advances, making coordinated activities possible. The baby begins to develop language skills.

From 2 years to puberty, you're looking at a *child*. Influenced by growth hormones, growth continues its rapid pace as deciduous teeth are replaced by permanent teeth. Muscle coordination, language skills, and intellectual skills also develop rapidly.

From puberty, which starts between the ages of 11 and 14, to adulthood, the child is called an *adolescent.* Growth occurs in spurts. Girls achieve their maximum growth rate between the ages of 10 and 13, whereas boys experience their fastest growth between the ages of 12 and 15. Primary and secondary sex characteristics begin to appear. Growth terminates when the epiphyseal plates of the long bones ossify sometime between the ages of 18 and 21 (see Chapter 7 for an introduction to the bones of the body). Motor skills and intellectual abilities continue to develop, and psychological changes occur as adulthood approaches.

The young adult stage covers roughly 20 years, from the age of 20 to about 40. Physical development reaches its peak and adult responsibilities are assumed, often including a career, marriage, and a family. By age 25, brain development is complete, marked by the maturation of the *prefrontal cortex* — the part of the brain responsible for *executive functions* like planning, decision making, and problem solving. After about age 30, physical changes that indicate the onset of aging begin to occur.

From age 40 to about 65, physiological aging continues. Gray hair, diminished physical abilities, skin wrinkles, and other outward signs of aging are caused by the decreased activity of cells and the lack of their replacement. Most notably, collagen and elastin fibers break down and are not replaced.

Women go through *menopause*, the cessation of monthly cycles. While *menarche* (the onset of menstruation) may begin any time between 10 and 15 years of age, the female body's monthly reproductive cycle slows and stops entirely between the ages of 40 and 55. With the cessation of menses comes a decrease in size of the uterus, shortening of the vagina, shrinkage of the mammary glands, disappearance of Graafian follicles, and shrinkage of the ovaries. For about six years prior to menopause, many women experience a stage called *perimenopause* during which increasingly irregular hormone secretions can cause fluctuations in menstruation, a sensation called *hot flashes*, and mood irritability. At around the same age, males may experience *andropause* due to a drop in testosterone. However, this drop is gradual and often does not completely cease, as estrogen does in women. Some men feel fatigued, irritable, and depressed, and they experience lack of sex drive, though others maintain their drive as well as the reproductive functionality (production of sperm) their whole life.

REMEMBER From age 65 until death is the period of *senescence*, the gradual decline in function we call aging. Individual adults can show widely varying patterns of aging in part because of differences in genetic background and physical activities. Signs of senescence include

- Loss of skin elasticity and accompanying sagging or wrinkling
- Weakened bones and decreasingly mobile joints
- Weakened muscles
- Impaired coordination, memory, and intellectual function
- Cardiovascular problems
- Reduced immune responses
- Decreased respiratory function caused by reduced lung elasticity
- Decreased peristalsis and muscle tone in the digestive and urinary tracts

Following are some practice questions on the aging cycle:

59 An individual's brain is considered fully developed

a. during prepubescence.
b. upon onset of menopause.
c. during adolescence.
d. during young adulthood.
e. on the onset of senescence.

60 What is the formal term for a 20-month-old child?

a. Baby
b. Infant
c. Toddler
d. Neonate
e. Child

61-67 Match the description to its life stage.

a. Neonate
b. Infant
c. Child
d. Adolescent
e. Young adult
f. Middle-aged adult
g. Old adult

61. _____ Assumes adult responsibilities, possibly including marriage and a family
62. _____ Faced with survival, it must process food, excrete waste, and obtain oxygen
63. _____ Primary and secondary sex characteristics begin to appear
64. _____ Experiences senescence
65. _____ Deciduous teeth begin to form
66. _____ Women go through menopause
67. _____ From 2 years of age to puberty

Answers to Questions on the Female Reproductive System

The following are answers to the practice questions presented in this chapter.

1 True or False: The medulla of the ovary is the site of follicular development. **False.** This occurs in the outer layer, the cortex; otherwise how would the egg get out?

2-11 Fill in the blanks to complete the paragraph.

The **2. oogonia** in a developing female fetus begin meiosis. Upwards of 2 million **3. primordial follicles** are produced but they are suspended in **4.**

prophase I. By the onset of **5. puberty** only about 300,000-400,000 remain. Each month, about 1,000 primary follicles are triggered to resume meiosis by **6. FSH.** One usually outgrows the rest forming the **7. Graafian follicle.** Meiosis is again suspended, this time at **8. Metaphase II**. Release of the secondary oocyte, or **9. ovulation,** is triggered by the release of **10. LH** from the anterior pituitary. Meiosis II is not completed until **11. fertilization** has begun.

12 Corona radiata: **c. Granulosa cells forming the outer layer of the Graafian follicle**

13 Endometrium: **e. Inner lining of the uterus**

14 Fimbriae: **a. Fingerlike projections at the end of a uterine tube**

15 Zona pelludica: **b. Protective layer directly surrounding the secondary oocyte**

16 Myometrium: **d. Layer of smooth muscle in the uterine wall**

17 Which one of the following is not a function of estradiol? **c. Supporting development of the corpus luteum.** By the time the corpus luteum starts to develop, estrogen already has begun to bow out of the reproductive equation, allowing progesterone to take the lead.

18-33 Following is how Figure 17-2, the female reproductive system, should be labeled.

18. **n. Uterine (Fallopian) tube;** 19. **f. Ovary;** 20. **l. Uterus;** 21. **c. Urinary bladder;** 22. **k. Mons pubis;** 23. **p. Urethra;** 24. **d. Clitoris;** 25. **i. Vaginal orifice;** 26. **m. Labium minor;** 27. **e. Labium major;** 28. **b. Fimbrae;** 29. **a. Cervix;** 30. **g. Rectum;** 31. **o. Vagina;** 32. **j. Anus;** 33. **h. Bartholin's glands**

34 Why does the corpus luteum produce progesterone? **c. To prepare a woman's uterus for pregnancy and prevent menstruation.** It stands to reason that if the corpus luteum forms after the Graafian follicle ruptures, the progesterone it's producing is signaling the uterus that a fertilized egg may be on its way.

35 True or False: Once a sperm penetrates the cell membrane of the ovum, the zona pelludica immediately hardens. **True:** This prevents another sperm from entering; 3 of each chromosome would likely mean the embryo would not develop.

36-44 Meiosis produces sperm and ova, which when combined make a **36. zygote** (fertilized egg) with its full complement of chromosomes. Normal humans have **37. diploid,** or 2N, cells containing 23 pairs of homologous chromosomes for a total of 46 chromosomes each. A pair of chromosomes containing the same type of genetic information are **38. homologous** chromosomes, though they may have different **39. alleles** or varieties of a gene. Ova and sperm are called sex cells or **40. gametes.** The number of chromosomes is cut in half during the first meiotic division, producing gametes that are **41. haploid,** or 1N. The second meiotic division produces **42. four haploid** sperm in the male but only **43. one functional haploid** secondary oocyte in the female. The zygote then proceeds through **44. mitosis** to produce the body's cells.

45 The mesoderm will form nervous tissue and skin. **False.**

46 The embryonic stage is complete at the end of the eighth week. **True.**

47 Outer cells of the embryonic disk form the germ layers. **False.** The inner cells do this; the outer cells form the amniotic sac.

48 During the fifth through seventh weeks, the arm and leg buds elongate, and fingers and toes begin to form. **True.**

49 Cleavage is successive mitotic divisions of the embryonic cells into smaller and smaller cells. **True.**

50 As the zygote moves through the uterine tube, it undergoes meiosis. **False.** Meiosis is completed during fertilization.

51 The placenta exchanges gases and waste between the maternal blood and the fetal blood. **True.**

52 After five days of cleavage, the cells form into a hollow ball called the morula. **False.** When the embryo hollows out it becomes the blastocyst.

53 Sexual intercourse five days before ovulation cannot lead to pregnancy. **False.** It certainly can happen — and does. As long as viable sperm are present when the egg arrives, conception can occur.

54 What needs to happen before a fetus can be considered viable? **a. All organ systems must be functional.** This generally occurs by the third trimester, but even then a premature birth can have serious health consequences for the newborn.

55 Describe one new fetal development for each month:

3rd month: Bones begin to ossify, body growth accelerates while head growth slows, lungs begin to practice breathing amniotic fluid, external genitalia visible in male, 4 inches long and about 1 ounce

4th month: Body grows rapidly, legs lengthen, joints begin to form, face looks more human, roughly 7 inches long and 4 ounces

5th month: Mother can feel fetal movement, hair grows on head, lanugo covers skin, about 12 inches long and ½ to 1 pound

6th month: Eyebrows and eyelashes form, weight gain accelerates, roughly 11 to 14 inches long and just under 1½ pounds

7th month: Subcutaneous fat begins to form, eyelids open, usually turns to upside-down position, between 13 and 17 inches long and 2½ to 3 pounds

8th month: Subcutaneous fat increases, fetus appears more "babylike," testes of male descend into scrotum, 16 to 18 inches long and roughly 5 pounds

9th month: Substantial "plumping" with subcutaneous fat, lanugo shed, fingernails extend to fingertips, average newborn is 20 inches long and weighs 7½ pounds

56 Which hormones play an active role in the progression of labor? **d. Prostaglandins and Oxytocin.** Progesterone blocks uterine contractions. Its decline at the end of pregnancy allows the labor.

57 Which stage of labor usually lasts the longest? **a. stage 1, phase 1.** While certainly not the most painful, early labor can last upwards of an entire day.

58 Explain how childbirth is an example of a positive feedback loop. **In negative feedback, the body's response to a stimulus is to push in the opposite direction; for example, the body sweats in response to a temperature increase in order to bring it down. In positive feedback, the body's response is to push further in the same direction. There are numerous players in this loop but the bottom line is: Contractions get you more contractions which, in turn, get you more contractions.**

59 An individual's brain is considered fully developed **d. during young adulthood.** This doesn't mean you lose the capacity to learn, just that your brain is done developing the parts.

60 What is the formal term for a 20-month-old child? **a. Infant.** The terms "baby" and "toddler" aren't part of the formal medical lexicon, and a neonate is less than 4 weeks old.

61 Assume adult responsibilities, possibly including marriage and a family: **e. Young adult**

62 Faced with survival, it must process food, excrete waste, and obtain oxygen: **a. Neonate**

63 Primary and secondary sex characteristics begin to appear: **d. Adolescent**

64 Experiences senescence: **g. Old adult**

65 Deciduous teeth begin to form: **a. Infant**

66 Women go through menopause: **f. Middle-aged adult**

67 From 2 years of age to puberty: **c. Child**

Part 6

The Part of Tens

IN THIS PART ...

Brush up on the ten best ways to study anatomy and physiology effectively.

Explore ten fun physiology facts in a free-wheeling testament to trivia.

Chapter 18

Ten Study Tips for Anatomy and Physiology Students

IN THIS CHAPTER

- **Playing with words and outlines**
- **Getting the most out of study groups**
- **Tackling tests and moving forward**

What's the best way to tackle anatomy and physiology and come out successful on the other side? Of course, a good memory helps plenty — after all, you have to remember what goes where and which terminology attaches to which part. But with a little advance planning and tricks of the study trade, even students who complain that they can't remember their own names on exam day can summon the right terminology and information from their scrambled synaptic pathways. In this chapter, we cover ten key things you can start doing today to ensure success not only in anatomy and physiology but in any number of other classes.

Writing Down Important Stuff in Your Own Words

This is a simple idea that far too few students practice regularly. Don't stop at underlining and highlighting important material in your textbooks and study guides: Write it down. Combine your notes with the class notes. Whatever you do, don't just regurgitate it exactly as presented in the material you're studying. Find your own words. Create your own analogies. Tell your own tale of what happens to the bolus as it ventures into the digestive tract. Detail the course followed by a molecule of oxygen as it enters through the nose. Draw pictures of the differences between meiosis and mitosis. When you're answering practice questions, pay special attention to the ones you get wrong.

Write reflections about why you answered incorrectly and what you need to remember about the right answer.

Unfortunately, typing the information doesn’t create as strong of a connection so get that pencil, pen, marker, or even crayon moving! Completely relax into the process with the knowledge that no one else ever has to see what you write or sketch. All anyone else will ever see is your successful completion of the course!

Gaining Better Knowledge through Mnemonics

Studying anatomy and physiology involves remembering lists of terms, functions, and processes. Sprinkled throughout this book are suggestions to take just the first letter or two of each word from a list to create an acronym. Occasionally, we help you go one step beyond the acronym to a clever little thing called a *mnemonic device*. Simply put, the mnemonic is the thing you commit to memory as a means for remembering the more technical thing for which it stands. For example, a question in Chapter 7 asks you to list in order the epidermal layers from the dermis outward; we suggest that you commit the following phrase to memory: Be Super Greedy, Less Caring. Just like that, a complicated list like basale, spinosum, granulosum, lucidum, corneum gets a little closer to a permanent home in your brain.

EXAMPLE Not feeling terribly clever at the moment you need a useful mnemonic? Surf on over to `www.medicalmnemonics.com`, which touts itself as the world’s database of these useful tools. Here’s a sampling of the site’s offerings:

- To remember the valves of the heart: “Try Pulling My Aorta: **T**ricuspid, **P**ulmonary, **M**itral, **A**ortic.”
- To remember the bones of the eye socket: “My Little Eye Sits in the orbit: **M**axilla, **L**acrimal, **E**thmoid, **S**phenoid.”
- To remember the upper arm muscles: “3 Bs bend the elbow: **B**iceps,

Brachialis, **B**rachioradialis."

- To remember the path out of the body from the top of the intestines: "Dow Jones Industrial Average Closing Stock Report: **D**uodenum, **J**ejunum, **I**leum, **A**ppendix, **C**olon, **S**igmoid, **R**ectum."

Discovering Your Learning Style

Every person has his or her own sense of style, and woe betide anyone who tries to shoehorn the masses into a single style. The same, of course, is true of students. To get the most out of your study time, you need to figure out what your learning style is and alter your study habits to accommodate it. No idea what we're talking about? Answer the questionnaire posted at `www.vark-learn.com/english/page.asp?p=questionnaire`, and the VARK guide to learning styles will tell you more about yourself than your last psychotherapist.

VARK, as you may have suspected, is an acronym. It stands for Visual (learning by seeing), Aural (learning by hearing), Reading/Writing (learning by reading and writing), and Kinesthetic (learning by touching, holding, or feeling). If you're a visual learner, you may get more out of anatomy and physiology by seeing the real thing in the flesh. If you're an aural learner, you may learn best in the classroom as the teacher lectures. If you're a reading and writing kind of learner, you'll get the most out of our first tip to write stuff down. And if you're a kinesthetic learner, there's nothing like touching or holding to commit something to memory.

TIP Study in ways that match your learning style. Visual learners don't just benefit from pictures but also converting information into flowcharts. Aural learners need to say everything out loud when practicing rather than just write it down. Reading/Writing learners just read, write, and repeat. Kinesthetic learners need to make their studying as active as possible. Most K learners benefit from visual-based study techniques, particularly flow charting and grouping information. It can be difficult to touch concepts, but something as simple as kicking a ball around when reviewing can be helpful. This also holds true for those learners who tend to drift off to la-la land easily.

Getting a Grip on Greek and Latin

TIP If you keep thinking "It's all Greek to me," congratulations on your insight! The truth of the matter is that most of it actually *is* Greek. So dust off your foreign language learning skills and begin with the basic vocabulary of medical terminology. Get started with the Greek and Latin roots, prefixes, and suffixes that appear on this book's Cheat Sheet at `www.dummies.com/cheatsheet/anatomyphysiologywb`. You'll soon discover that for every "little" word you learn, a whole mountain of additional terms and phrases are just waiting to be discovered.

Connecting with Concepts

It happens time and again in anatomy and physiology: One concept or connection mirrors another yet to be learned. But because you're focusing so hard on this week's lesson, you lose sight of the value in the previous month's lessons. For example, a concept like metabolism comes up in a variety of ways throughout your study of anatomy and physiology. When you encounter a repeat concept like that, create a special page or two for it at the back of your notebook or link the concept to a separate computer file. Then, every time the term comes up in class or in your textbook, add to the running list of notes on that concept. You'll have references to metabolism at each point it comes up *and* you'll be able to analyze its influences across different body systems.

Forming a Study Group

If you're really lucky, someone in your class (or maybe it's even you) has already suggested forming that time-honored tradition — a *study group*. The power of group members to fill gaps in your knowledge is priceless. But don't restrict it to late-night cramming just before each test. Meet with your group at least once a week to go over lecture notes and textbook readings. If it’s true that people only retain about 10 percent of what they hear or read, then it makes sense that your fellow group members will recall things that slipped immediately from your mind.

Outlining What’s to Come

As you read through a chapter of your textbook to prepare for the next lecture, prepare an outline of what you’re reading, leaving plenty of space between subheadings. Then, during the lecture, take your notes within the outline you’ve already created. Piecing together an incomplete puzzle shows you where the key gaps in your knowledge may be. Additionally, familiarizing yourself with the upcoming vocabulary will lead to easier understanding when the concepts (especially those detailed processes) are presented.

Putting In Time to Practice

REMEMBER Ten minutes a day goes a long way! Flashcards, mnemonic drills, practice tests — be creative and practice, practice, practice! Sometimes instructors share tidbits about what they plan to emphasize, but sometimes they don’t. In the end, if you’ve done the work and put in the time to study and practice with information outside of class, the exact structure and content of an exam shouldn’t make much difference. Nailing down the labeling and vocabulary early leaves you more time to work with the complicated physiology as you approach that all important test.

Flashcards are a highly effective strategy, but only if you do it right. Before starting, map out your timeline: When do you need to have all these terms down, and how often will you review them? For example, if you have one week, then you could review twice per day. During the first run-through, place the cards into two piles: correct and incorrect. The incorrect pile stays in the first stack and the correct pile moves up to the once-a-day stack. Later, pick up that first stack and repeat; you get to ignore the other one. For the next day, start with the once-daily pile and make the correct/incorrect piles like always. Correct ones get moved up another notch (every two days); incorrect ones goes back down to twice daily. Then go through the twice daily stack. This way, you don't waste valuable study time on terms you already have committed to memory.

Sleuthing Out Clues

Okay, it's test time! Take advantage of the test itself. You may find that the answer to an exam question that stumps you is revealed — at least partially — in the phrasing of a subsequent question. Make skipping questions part of your test-taking routine. (Do you even have one? Or do you just sit down and take the test?) Don't guess until you have to. That way, you will be more alert to these blessed little gifts even when you think that you already understand all the anatomical structures and physiological processes.

Reviewing Your Mistakes

The test is done and the grades are in. So there was a really tough question on the test and you blew it big-time? Don't beat yourself up over it. It's hardly a missed opportunity — this is where rolling with the punches really pays off. Go back over the entire test and pay extra attention to what you got wrong. Start your next practice sessions with those questions, and stay alert for upcoming material that may trip you up in a similar way. Remember that learning from a mistake leads to a "stronger" memory than having memorized correctly in the first place. You're not likely to make that mistake again come final exam time.

Chapter 19

Ten Fun Physiology Facts

IN THIS CHAPTER

- **Boning up on physiology trivia**
- **Making discoveries about the brain, the skin, and more**

Basic anatomy may be fairly straightforward, but how the human body uses all those parts can present a smorgasbord of interesting discoveries. In this chapter, we give you just a peek at ten of the more intriguing aspects of what makes the body tick.

Boning Up on the Skeleton

Make no bones about it — the human skeleton is a trove of trivia. Try a few of these on for size (and check out Chapter 7 for the scoop on the skeleton):

- People are born with 300 bones in their infant bodies, but by the time they're adults they only have 206. Why? Because human bodies spend infancy and childhood putting the finishing touches on their skeletons, knitting together two or more bones into one. The only bone fully grown at birth is in the ear (the stapes). Plus, over a period of about seven years, each bone in the body is slowly replaced until it is a new bone.
- You're taller in the morning. No, really, you are. After you stand up from bed and get your day going, the cartilage between your back bones becomes compressed over time. By the end of the day, you're about one centimeter shorter.
- Bones make up about 14 percent of an average individual's weight, yet they're as strong as granite. A block of bone the size of a matchbox can support 9 tons — four times more weight than concrete.
- Bones can self-destruct. Excessive exposure to cadmium can prompt them to do so by triggering premature *apoptosis*, the controlled death of cells

that takes place as part of normal growth and development. Alternatively, if you're not taking in enough calcium, certain hormones can leach calcium from your own bones to try to balance the blood's supply of this essential mineral.

Flexing Your Muscles

Besides your heart, the strongest muscle in the body (compared to its size) is your tongue. Hey, something has to counterbalance those 200 pounds of force the jaw muscles can deliver when you're chewing. And which muscles move more each day than even your heart? The ones surrounding your eyes, called the *extraocular muscles*. At three moves per second, they clock in more than 100,000 motions per day.

Every time you take a step, you employ around 200 of the roughly 600 muscles in your body. Researchers estimate that every minute you spend walking extends your life by anywhere from 90 seconds to two minutes. No wonder doctors advise that people shoot for 10,000 steps each day!

Thinking you'll just lift weights instead? Choose free weights over using a weight machine. The action of balancing the weights yourself rather than letting a machine do it for you builds muscle mass faster. (Chapter 8 introduces you to the muscles.)

Fighting Biological Invaders

As the body's internal defense network, the immune system sets blood and lymph against biological invaders seeking to colonize you. But its first line of defense is also the body's largest organ: the skin. If your skin — and some of the "friendly" bacteria living there — didn't secrete natural antibacterial compounds, you could wake up in the morning coated in microbial slime. As it is, though, most of the pathogens that land on you die quickly. (See Chapter 6 for details on the skin.)

The skin isn't the only thing working against the invading microbial hordes; the enzyme lysozyme found in human tears, saliva, and mucus is custom-made to disintegrate bacterial cell walls. (Remember: As eukaryotes, people's cells don't have walls.)

Not that all microorganisms are bad. You have between 2 and 5 pounds of bacteria living inside you, much of it in the intestines. As scientists have begun to understand what that microbial life is up to, it has become clear that your internal “microbiome” is a big part of what keeps you healthy. An imbalance in your gut flora could obviously lead to digestive issues but also vitamin deficiencies and may even influence mental health.

Cells Hair, There, and Everywhere

Every human alive today spent about 30 minutes at the start as a single cell. Now, however, your body is making 25 million new cells every day, and you shed and regrow all your outer skin cells about every 27 days. (See Chapter 6 for more about the skin.) Beards are the fastest-growing hairs on the human body; but all hairs have a genetically predetermined maximum length. The average human scalp has 100,000 hairs — blondes have more hair, on average, than dark-haired people do — and you lose between 40 and 100 strands of hair each day. Fingernails grow nearly four times faster than toenails do.

That’s quite a bit of cellular production going on. Ever wonder what happens to those cells when they die? Well, they don’t turn to dust. That’s a commonly held belief that is actually just a myth. (Dust is mostly good, old-fashioned dirt.) While a small percentage could certainly be dead skin cells, we lose most of them when we bathe. Another common myth is that your hair and nails continue growing after death. Growth does appear to take place, but it’s caused by the skin losing its bulk as the water evaporates. So the skin shrinks back, making it look like the hair or nails are growing out.

Swallowing Some Facts about Saliva and the Stomach

Besides being a digestive kick-starter, saliva prevents tooth decay and keeps your throat and mouth from drying out. It may not feel like it, but those six little glands make nearly half a gallon — about 1.5 liters — of saliva every day. Over the course of a lifetime, that’s enough to fill about two average-sized swimming pools.

Hard to stomach? Consider this: With hydrochloric acid inside that's so corrosive it can dissolve wood and steel, your stomach should consume itself. But it doesn't. Why? Because you make your own natural supply of antacid. The epithelial cells lining your stomach secrete a steady supply of bicarbonate that neutralizes stomach acid on contact. Plus, these cells have a short life span (about three days) so the lining is continuously being replaced.

Be forewarned, though: That soothing bicarbonate produces carbon dioxide, among other things, as a by-product. That has to go somewhere. Some of it comes up in the form of a belch. And some of it contributes to the 17 ounces per day of flatulence that the average healthy human releases. (See Chapter 14 for an introduction to the digestive system.)

Appreciating the Extent of the Cardiovascular System

We tell you in Chapter 11 about the heart, the hardest-working muscle in the entire body, and its web of liquid connective tissue more commonly known as blood. But we don't mention a few facts that may make you appreciate your cardiovascular system even more.

In one day, the average individual's heart exerts enough power to lift a 1-ton weight more than 40 feet off the ground. During that day's pumping, the body's red blood cells travel about 12,000 miles, or roughly half the distance around the Earth at the equator and they complete a full lap every 60 seconds. They've got plenty of room to roam; if you laid every blood vessel in a body end-to-end, they would stretch around the Earth more than two times!

If that's hard to believe, consider this: Every square inch of skin contains 20 feet of blood vessels. Tissue the size of a pinhead contains 2,000 to 3,000 capillaries. You certainly have plenty of blood to fill all that space; you have 2.5 trillion red blood cells, more or less, in your body at any given moment, and your bone marrow creates 100 billion new ones every single day. That's so much blood that it would take 1.2 million mosquitoes each drinking their fill once to completely drain the average human of blood.

"You're Glowing" Isn't Just an

Expression

You may have heard the expression before when you just got a raise, found out you're pregnant, or something awesome has happened and people say, "Look at you. You're glowing!" Turns out, that's not just an expression — you very well might be glowing. Scientists confirmed in 2009 that humans generate *bioluminescence*, just not at an intensity that our eyes can pick up on. Photons, or particles of light, are generated by excited molecules from cellular respiration. The glow is most prevalent in the late afternoon around the forehead, neck, and cheeks.

Another feature we can't see are our stripes. Called *Blaschko's lines*, these stripes are generated by the normal embryonic development of skin and are the same pattern for everyone. Most of us will never see them, but some skin conditions will follow the lines, making them more evident.

Looking at a Few of Your Extra Parts

With so many industrious components keeping you moving through your life, it can be startling to think about the number of body parts that, frankly, you just don't need. The first ones that come to mind often are the appendix and the wisdom teeth. The appendix, which does produce a few white blood cells and house important gut bacteria, generally gets to stay put so long as it doesn't get infected. But that third set of molars does little but cause pain and push around the teeth you really need.

You also have a "tailbone," or coccyx, which serves no function other than as a high-scoring Scrabble word, and it's a painful reminder of hard falls on your backside. It's a collection of fused vertebrae that researchers believe are what's left of the mammalian tail our evolutionary ancestors once sported. Other features that harken back to our early ancestors are the arrector pili that give us goosebumps (that don't actually keep us warm) and extrinsic ear muscles (though a small portion of people maintain control of these and can do a pretty neat ear wiggle).

People have a few other vestigial organs. Men, believe it or not, have an undeveloped vestigial uterus hanging on one side of the prostate gland, while women have a vestigial vas deferens — a cluster of tubules near the ovaries that would have become sperm ducts if they had inherited a Y chromosome from their fathers. (Part 5 has more details on both the male and female reproductive systems.)

Understanding Your Brain on Sleep

Superlatives about the human body tend to center around the brain. Fastest, neediest, most powerful — it is, after all, what makes us human. That 3-pound hunk of tissue demands 20 percent of the oxygen and calories the body takes in, communicates with 45 miles of nerves in the skin, contains individual neurons that can live more than a century, and generates enough energy when a person is awake — between 10 and 23 watts — to power a light bulb.

But the fun really begins when humans go to sleep. Scientists long have wondered why the body requires that people spend one-third of their lives unconscious. Although an individual may look quiet when she's with the sand man, studies have shown that the brain is more active in sleep than awake. Sleep, and the process of dreaming, helps the brain consolidate memories in ways that researchers are only beginning to understand.

For most of us, communication with the cerebellum is shut off by the pons so that we don't physically make the movements we are making in our dreams. But if you know anyone with *parasomnia* (sleep walking, sleep talking, and so on), you know this switch doesn't always work — likely because it's way more complicated than a single "switch."

Getting Sensational News

When people talk about sensation, most of the time they're referring to the five primary senses — vision, hearing, taste, touch, and smell (see Chapter 9). But the notion of a "sixth sense" may be more than the stuff of science fiction. Vision combines senses for both light and color, and there is growing evidence that your *proprioception* — the ability to detect your relative position in space — may rely on your body's ability to detect magnetic fields in much the same way migrating birds do. Blind people have been known to develop echolocation, or the ability to hear subtle changes in sound bouncing back from otherwise unseen objects. And stress sometimes causes people to experience time dilation.

In fact, your senses are more subjective than you may like to admit. Things get even more complex when you consider the condition known as *synesthesia*, in which a person can “hear” color or “see” sound. The eyes, which can distinguish up to one million colors, routinely take in more information than the largest telescope ever created. The nose can identify and the brain subsequently can remember more than 50,000 smells. Static touch can discern an object about twice the diameter of an eyelash, while dynamic touch — dragging a finger along a surface — can detect bumps the size of a very large molecule.

About the Authors

Erin Odya is an anatomy & physiology teacher at Carmel High School in Carmel, Indiana, which is widely regarded as one of Indiana's top schools. She takes great pride in her role as an educator. Erin is also the author of *Anatomy & Physiology for Dummies*, 3rd Edition.

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Dedication

To Gabe and his never-ending questions about how his 8-year-old body works and to all of my students — past, present, and future — who challenge my knowledge and inspire my curiosity. —Erin Odya

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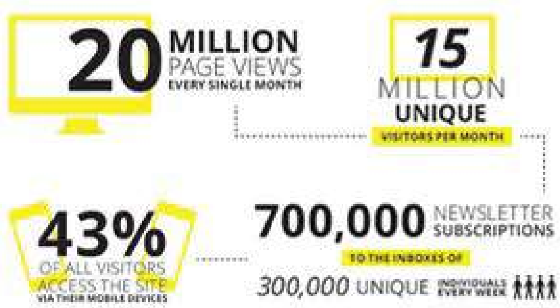

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